ARANYA SAHA

Cell: +1 240-960-7398 Email: aranya@umd.edu LinkedIn: Aranya Saha

EDUCATION

University of Maryland, College Park Aug 2026 – Present
PhD in Electrical and Computer Engineering
Research Advisor: *Prof. Sanghamitra Dutta*

Bangladesh University of Engineering and Technology Feb 2020 – Mar 2025
BSc in Electrical and Electronic Engineering
Research Advisor: *Prof. Mohammad Ariful Haque*

WORK EXPERIENCE

University of Maryland, College Park Aug 2026 – Present
Graduate Research Assistant *MD, USA*

Advanced Chemical Industries PLC Apr 2025 – Aug 2026
Machine Learning Engineer *Dhaka, Bangladesh*

Bangladesh University of Engineering and Technology Nov 2025 – Apr 2026
Part-time Lecturer, Electrical and Electronic Engineering *Dhaka, Bangladesh*

Institute of Robotics and Automation, BUET Jun 2025
Instructor, Robotics Bootcamp 2025 *Dhaka, Bangladesh*

RESEARCH INTERESTS

Efficient AI, Mechanistic Interpretability, Reinforcement Learning, Large Language Models, AI for Science.

PUBLICATIONS

Preprints

1. Ismam Nur Swapnil, **Aranya Saha**, Tanvir Ahmed Khan, Mohammad Ariful Haque, Ser-Nam Lim, "Gradient Extrapolation-Based Policy Optimization." – Link
2. **Aranya Saha***, Tanvir Ahmed Khan*, Ismam Nur Swapnil*, Mohammad Ariful Haque, "CLARIFY: A Specialist–Generalist Framework for Accurate and Lightweight Dermatological Visual Question Answering." – Link
3. Ismam Nur Swapnil*, **Aranya Saha***, Tanvir Ahmed Khan*, Mohammad Ariful Haque, "GRPO++: Enhancing Dermatological Reasoning under Low Resource Settings." – Link
4. Tanvir Ahmed Khan, **Aranya Saha**, Ismam Nur Swapnil, Mohammad Ariful Haque, "Compression Strategies for Efficient Multimodal LLMs in Medical Contexts." – Link

Conferences

1. Shadman Sobhan, **Aranya Saha**, Tanvir Ahmed Khan, Abduz Zami, "Skin Cancer Classification Using Pretrained CNNs: A Transfer Learning Approach Addressing Imbalanced Data Challenges," International Conference on Next-Generation Computing, IoT and Machine Learning (2025). – Link
2. Shadman Sobhan, Abduz Zami, Mohiuddin Ahmed, Tanvir Mahtab Zihan, Tanvir Ahmed Khan, **Aranya Saha**, "A Multi-Stage Deep Learning Approach to Tuberculosis Detection with Explainable Insights," International Conference on Next-Generation Computing, IoT and Machine Learning (2025). – Link

SELECTED PROJECTS

1. **Efficient Frame Selection for Long Egocentric Video Understanding.** Developed an efficient frame selection approach for long-form egocentric video understanding. – GitHub
2. **EchoLens: Multimodal Conversational AI Engine.** Developed a multimodal conversational AI system integrating vision and language models for interactive understanding and response generation. – GitHub
3. **Simple MedQA-GPT: GPT Tailored for Medical Q&A.** Developed a GPT-based system for answering medical questions with domain-specific prompting and conversational interaction. – GitHub
4. **IoT-Based Patient Health Monitoring System.** Developed an IoT-enabled system for real-time monitoring of patient health parameters. – GitHub Report
5. **MATLAB-Based Fingerprint Recognition System.** Developed a fingerprint recognition system using MATLAB for biometric identification and matching. – GitHub Report

RELEVANT COURSEWORK

Graduate	Random Processes, Advanced Digital Signal Processing
Undergraduate	Artificial Intelligence and Machine Learning, Digital Signal Processing, Random Signals and Processes, Probability and Statistics, Signals and Systems, etc.

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB
ML/DL	PyTorch, TensorFlow, CNNs, VLMs, LLMs
Data/Tools	NumPy, Pandas, Git, Docker, FastAPI, LaTeX
Hardware	Microcontrollers, IoT Devices, Sensors

LEADERSHIP EXPERIENCE

Association for Computing Machinery (ACM) – SIGCOMM Apr 2024 – Feb 2025
Student Executive *Remote*

Served as the first Student Executive of ACM SIGCOMM, working alongside Prof. Matthew Caesar (UIUC), Chair of ACM SIGCOMM; developed its official website under his mentorship and co-established a paper reading group.

HONORS AND AWARDS

- Dean's Fellowship (ECE) – University of Maryland, College Park (Fall 2026).
- **1st Runner-Up, Poster Competition (AI)** – BEAR Summit 2025, Poster Link, Certificate Link
- University Merit Scholarship – BUET, awarded for academic performance (4 semesters).
- Dean's List Award – BUET, awarded for academic performance (Years 1–2).
- Ranked 29th among 10,000+ candidates in the BUET Undergraduate Admission Test (2019).
- Ranked 31st (Male) in the Dhaka Board HSC Examination; recipient of the Talent Pool Scholarship.
- Perfect Attendance Certificate – Notre Dame College.

REFERENCES

Prof. Sanghamitra Dutta
Electrical and Computer Engineering
University of Maryland, College Park
Email: sanghamd@umd.edu

Prof. Mohammad Ariful Haque
Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Email: arifulhoque@eee.buet.ac.bd